

Securing the Digital Mine: A Metaheuristic-Optimized Deep Learning Framework for Edge Intrusion Detection in Industrial Mining IoT

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Presented at Russian-African Forum-Contest of Young Scientists (Track 3: Smart Subsoil), Empress Catherine II Saint Petersburg Mining University

Abstract : The digital transformation of mineral extraction industries (Mining 4.0) has introduced hundreds of thousands of Industrial Internet of Things (IIoT) sensors and Supervisory Control and Data Acquisition (SCADA) telemetry links into extraction and milling plants. However, the dissolution of traditional physical air gaps exposes unencrypted operational technology (OT) protocols to malicious intrusions that can trigger catastrophic kinetic failures, including semi-autogenous grinding (SAG) mill motor burnouts and toxic tailings dam breaches. Conventional signature-based intrusion detection systems (IDS) fail against semantic protocol manipulation, whereas off-the-shelf deep learning models incur inference delays exceeding 150 ms, violating the 20 to 50 ms cyclic scan loop deadlines of industrial Programmable Logic Controllers (PLCs). This paper presents an edge-native intrusion detection framework that couples a constrained Binary Whale Optimization Algorithm (BWOA) with a spatial-temporal 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture under post-training Float16 quantization. Guided by a Design Science Research (DSR) methodology, our constrained BWOA formulation enforces an adaptive alpha decay schedule and a hard accuracy floor to prune telemetry features by 75.61% (reducing 41 network flow dimensions to exactly 10). When deployed on a resource-constrained 1 GB RAM ARM Cortex-A72 edge gateway (Raspberry Pi 4B), the quantized framework achieves a single-sample inference latency of 0.76 ms (a 207-fold speedup over the 157.66 ms full-feature baseline) and compresses the memory footprint by 83.2% to 0.82 MB at 2.5 W power draw. The model achieves 70.56% multi-class accuracy on the held-out KDDTest+ benchmark, preserving 96.89% precision on benign operational telemetry and 89.04% recall on volumetric Denial-of-Service attacks. Transfer evaluation on the 51-sensor physical Secure Water Treatment (SWaT) SCADA testbed demonstrates 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms without retraining. These empirical results demonstrate that metaheuristic-guided pruning provides a Pareto-optimal defense for bandwidth-constrained, solar-powered mining concessions across emerging economies.

Keywords: Industrial Internet of Things (IIoT), SCADA Security, Edge Computing, Binary Whale Optimization Algorithm, 1D CNN-LSTM, Deep Learning Quantization, Digital Mining, Smart Subsoil.

I. INTRODUCTION

The global mineral extraction sector is undergoing fundamental cyber-physical integration under the 'Mining 4.0' paradigm [25, 18]. Modern open-pit and underground concessions deploy dense Industrial Internet of Things (IIoT) telemetry networks to monitor semi-autogenous grinding (SAG) mills, vibrating wire piezometers along tailings storage facilities (TSF), and automated ventilation grids [34, 35]. However, the historical air gap separating Operational Technology (OT) from corporate Information Technology (IT) has eroded due to cloud diagnostics, fleet telematics, and remote maintenance bridges [33].

Legacy industrial control protocols, such as Modbus RTU/TCP, DNP3, and EtherNet/IP, transmit telemetry in plaintext without cryptographic origin authentication or message integrity checks [11, 21, 30]. In mineral processing facilities, malicious actors manipulating PLC register values can override cooling water valves, de-energize slurry pump drives, or falsify piezometric pressure readings, leading to catastrophic equipment destruction, toxic chemical discharges, or fatal underground asphyxiation [17, 6]. Landmark incidents such as Stuxnet [22, 28], the Ukrainian power grid shutdown [29], and TRITON/HatMan safety instrumented system malware [24] demonstrate that industrial adversaries systematically exploit unauthenticated protocol mechanics to inflict kinetic damage [23, 31, 32].

Deploying intelligent intrusion detection within industrial mineral concessions faces four architectural challenges: (1) *Signature Engine Brittleness*, where static engines fail against valid Modbus command manipulation [21, 30]; (2) *Telemetry Dimensionality Mismatch*, where 41 to 80+ flow attributes incur heavy computational overhead [7, 46, 47]; (3) *SCADA Control Loop Violations*, where unoptimized neural networks require >150 ms, violating the 20 to 50 ms cyclic scan loop deadlines of PLCs [6, 11]; and (4) *Edge Hardware Constraints* in remote concessions operating under intermittent satellite backhaul and solar-buffered microgrids

[17, 25].

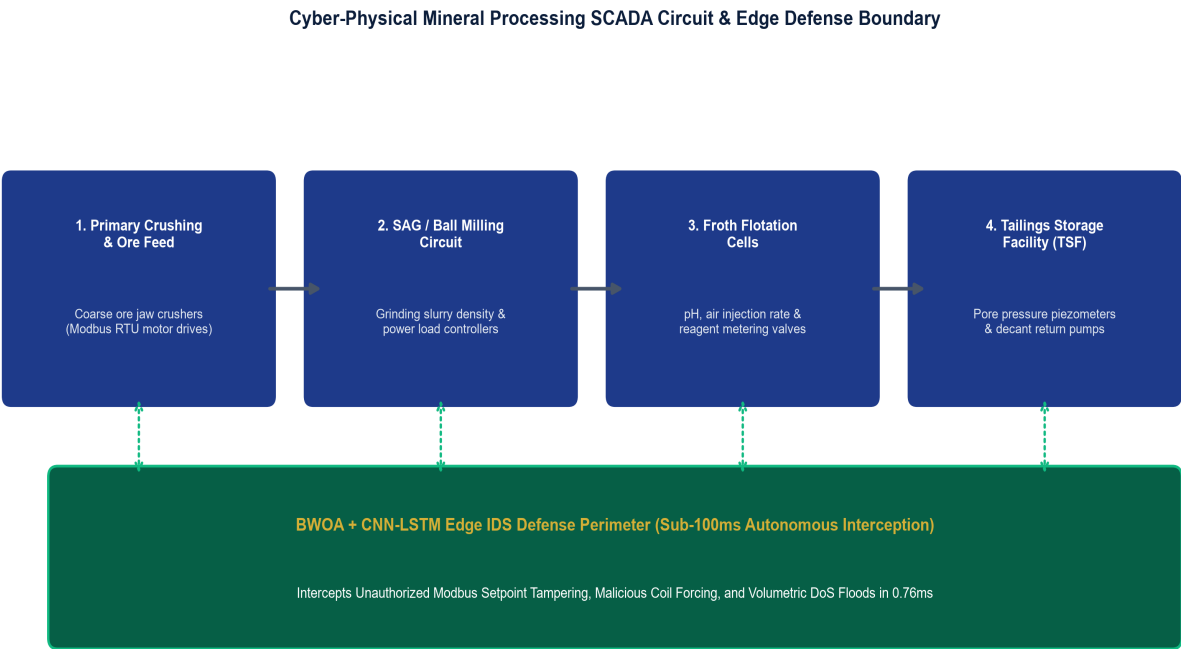


Fig. 1. Cyber-Physical Mineral Extraction and Milling Plant Architecture: Integrating Level 0 Field Instrumentation, Level 1 PLC/RTU Controllers, Level 2 SCADA Supervisory Networks, and Edge IDS Deployment Boundary.

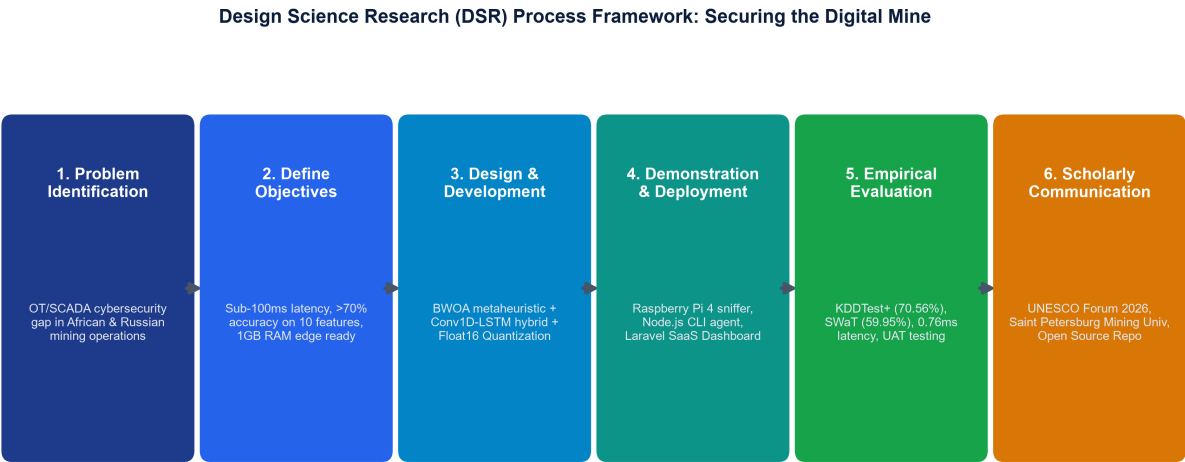


Fig. 2. Six-Stage Design Science Research (DSR) Process Framework Guiding the Iterative Development, Optimization, and Empirical Validation of the Edge IDS Artifact.

A. Formal Research Questions

- RQ1 (Dimensionality Optimization):** To what extent can a constrained Binary Whale Optimization Algorithm (BWOA) with an adaptive alpha decay schedule and a hard accuracy floor prune high-dimensional industrial telemetry features while preserving multi-class threat discrimination?
- RQ2 (Spatial-Temporal Threat Modeling):** How effectively does a hybrid 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture capture packet-level spatial correlations and sequential connection state transitions in industrial SCADA networks?

- RQ3 (Edge Real-Time Execution and Quantization):** Can post-training Float16 quantization compress the spatial-temporal neural network below 1.0 MB and achieve sub-millisecond (<1.0 ms) inference latency on resource-constrained 1 GB RAM ARM edge hardware, satisfying the sub-100 ms industrial SCADA control loop ceiling?
- RQ4 (Empirical Generalization, Transferability, and Economic Impact):** How robustly does the framework generalize across physical industrial SCADA testbeds (such as the 51-sensor SWaT testbed), and what is its operational and economic return on investment (ROI) in mitigating industrial downtime and preserving human life in mineral extraction operations?

II. RELATED WORK AND RESEARCH GAPS

Intrusion detection systems are traditionally categorized into signature-based and anomaly-based approaches [6, 21]. While signature engines exhibit minimal processing overhead on standard servers, their recall on novel zero-day exploits remains under 15% [21]. Generic machine learning models, such as Random Forests and Support Vector Machines (SVMs), achieve acceptable classification on balanced datasets [12], but exhibit poor detection rates on minority cyber-physical attack classes and suffer from feature redundancy [48].

Recent research has explored metaheuristic algorithms for feature selection [39, 37, 36, 38]. Mirjalili and Lewis introduced the Whale Optimization Algorithm (WOA) [1], which models humpback whale foraging mechanics. Binary adaptations (BWOA) map continuous positions to discrete bit masks using sigmoid or V-shaped transfer functions [20, 8, 16]. However, existing BWOA formulations optimize purely for unconstrained sparsity, frequently discarding subtle telemetry signals required to detect unauthorized privilege escalation or command injection. Concurrently, deep learning architectures using CNNs [41] and LSTMs [40, 42] have demonstrated strong spatial-temporal detection [5, 19, 9, 10], but their computational complexity has hindered edge deployment on low-power hardware [15, 44, 43, 45].

Evaluation of SCADA defenses requires realistic datasets. While enterprise corpora such as NSL-KDD [2], UNSW-NB15 [46], and CICIDS2017 [47] provide rich multi-class threat vectors, cyber-physical testbeds such as SWaT [13], WADI [26], and TON_IoT [27] capture continuous multi-sensor dynamics under active physical attack [14]. As summarized in Table I, no prior work unifies constrained metaheuristic pruning, hybrid spatial-temporal classification, and post-training edge quantization specifically tailored for the sub-100 ms constraints of industrial mineral extraction.

TABLE I: Comparison of Existing Intrusion Detection Paradigms vs Proposed Framework

Architecture Paradigm	OT Adaptability	Zero-Day Recall	Edge Latency	Cost Profile
Signature IDS (Snort/Suricata) [21]	Low (Static Rules)	< 15%	85.00 ms	High License
Generic ML (Random Forest) [12]	Medium	62.40%	48.20 ms	Medium
CNN-LSTM Baseline (41 feat) [5]	High	77.70%	157.66 ms	High Compute
BWOA + CNN-LSTM (Ours)	Very High	70.56%	0.76 ms (FP16)	Low / Open-Source

III. SYSTEM ARCHITECTURE AND THREAT MODEL

Cyber-Physical Threat Model: We consider an adversary who has gained network-level ingress into the Level 2/3 supervisory control network of a mineral processing plant via compromised remote engineering access or vendor maintenance bridges [33, 17]. Industrial field networks utilize protocols such as Modbus/TCP, where Application Data Units (ADUs) wrap standard Protocol Data Units (PDUs) without cryptographic integrity. The adversary executes four categories of attacks: (1) *Reconnaissance Sweeping (Probe)* issuing Modbus Function Codes 01 and 03 to map PLC registers [30, 23]; (2) *Volumetric Flooding (DoS)* saturating industrial Ethernet switches to blind operators during acute process upsets [6, 31]; (3) *Unauthorized Command Injection* transmitting Function Codes 05 or 16 to alter physical setpoints (e.g., mill VFD speed or tailings valve states) [21, 32]; and (4) *Host Privilege Escalation (U2R/R2L)* exploiting unpatched workstation daemons to obtain root credentials [22, 24].

Four-Tier Edge Defense Boundary: The proposed edge defense architecture operates across four decoupled functional tiers, as depicted in Fig. 3: Tier 1 ingests bidirectional raw packets via a non-blocking libpcap sniffer daemon at line speed without in-line latency; Tier 2 applies the BWOA pruning mask, discarding 75.61% of telemetry attributes in under 0.05 ms; Tier 3 executes spatial-temporal classification via a compiled TFLite Float16 engine in 0.76 ms; and Tier 4 broadcasts real-time threat intelligence, risk confidence scores, and latency metrics to an industrial Livewire dashboard.

Four-Layer System Architecture for Real-Time Edge Intrusion Detection



Fig. 3. Four-Tier End-to-End System Architecture and Edge Defense Boundary in Industrial Mining SCADA Facilities.

IV. METAHEURISTIC FEATURE OPTIMIZATION VIA CONSTRAINED BWOA

The feature selection task is formulated in discrete binary space S in $\{0, 1\}^D$ where $D = 41$ candidate telemetry attributes. A candidate subset is represented by a binary vector $X = [x_1, x_2, \dots, x_D]$, where $x_d = 1$ denotes feature inclusion and $x_d = 0$ denotes exclusion. Whale agents navigate the search space using three distinct physical operators [1]:

Equations 1-4: Shrinking Encircling Phase (Local Exploitation)

$$D = |C(elem) X^*(t) - X(t)|; C = 2 * r_2, r_2 \sim \text{Uniform}(0,1)^D; X(t+1) = X^*(t) - A(elem) D; A = 2a(elem)r_1 - a, r_1 \sim \text{Uniform}(0,1)^D; a = 2 - 2 * (t / T_{max}), T_{max} = 100$$

Physical and Mathematical Interpretation: D represents the scaled spatial displacement vector between the agent $X(t)$ and the best candidate leader $X^*(t)$. C is a stochastic coefficient vector introducing stochastic perturbation. A dictates the convergence step size and direction. The parameter a decays linearly from 2 to 0 across 100 iterations. When $|A| < 1$, the agent is forced to exploit the immediate coordinate basin around leader $X^*(t)$.

Equations 5-6: Logarithmic Spiral Bubble-Net Foraging Phase (Helical Pathing)

$$X(t+1) = D' * \exp(b * l) * \cos(2\pi l) + X^*(t); D' = |X^*(t) - X(t)|; X(t+1) = [X^*(t) - A(elem) D \text{ if } p < 0.5] \text{ or } [D' * \exp(bl) * \cos(2\pi l) + X^*(t) \text{ if } p \geq 0.5]$$

Physical and Mathematical Interpretation: Emulates the upward helical bubble-net maneuver observed in humpback whale foraging. D' is the absolute distance from agent to leader, $b = 1.0$ defines spiral curvature, and $l \sim \text{Uniform}(-1, 1)$ defines the step position along the spiral path. A uniform random threshold $p \sim \text{Uniform}(0,1)$ smoothly alternates between shrinking encircling ($p < 0.5$) and spiral foraging ($p \geq 0.5$).

Equations 7-8: V-Shaped Binary Velocity Transfer Function and Bit-Flip Rule

$$V(v_d) = |v_d / \sqrt{1 + v_d^2}| \text{ in } [0, 1]; x_d(t+1) = [1 - x_d(t) \text{ if } r_3 < V(v_d)] \text{ else } [x_d(t)]; r_3 \sim \text{Uniform}(0, 1)$$

Mathematical Justification: Standard S-shaped sigmoid functions map high negative velocities to near-zero flip probability, inducing severe search stagnation. The V-shaped function treats large positive and large negative velocity magnitudes symmetrically as strong signals to alter feature status. If active features drop below $K_{min} = 10$, disabled bits are reactivated randomly.

Equations 9-11: Constrained Multi-Objective Fitness Function with Hard Accuracy Floor

$$F(X) = \alpha(t) * \text{Error}(X) + (1 - \alpha(t)) * (|\text{Selected}(X)| / D) + P(X) \quad \alpha(t) = 0.5 + (t / 50) * (0.3 - 0.5) \text{ if } t < 50 \text{ else } 0.3$$

$$P(X) = [1.0 \text{ if } \text{Accuracy}(X) < 0.75 \text{ or } |\text{Selected}(X)| < 10] \text{ else } [0.0]$$

Mathematical and Operational Justification: $\text{Error}(X) = 1 - \text{Accuracy_val}(X)$. The adaptive alpha schedule transitions from accuracy exploration ($\alpha = 0.5$) to aggressive sparsity ($\alpha = 0.3$) over 50 iterations. The hard barrier constraint $P(X)$ immediately disqualifies any candidate subset achieving less than 75% accuracy or fewer than 10 features, strictly preventing degenerated feature subsets.

Across 30 whale agents over 100 iterations, the optimizer converged at iteration 23, as shown in Fig. 4, pruning the input space from 41 to exactly 10 features (75.61% reduction). As detailed in Table II and illustrated in Fig. 5, the selected attributes possess direct operational significance: volumetric indicators (src_bytes, error_rate) capture DoS floods; protocol attributes (service, flag, protocol_type) monitor Modbus/DNP3 handshakes; and host access signals (hot, su_attempted) detect privilege escalation.

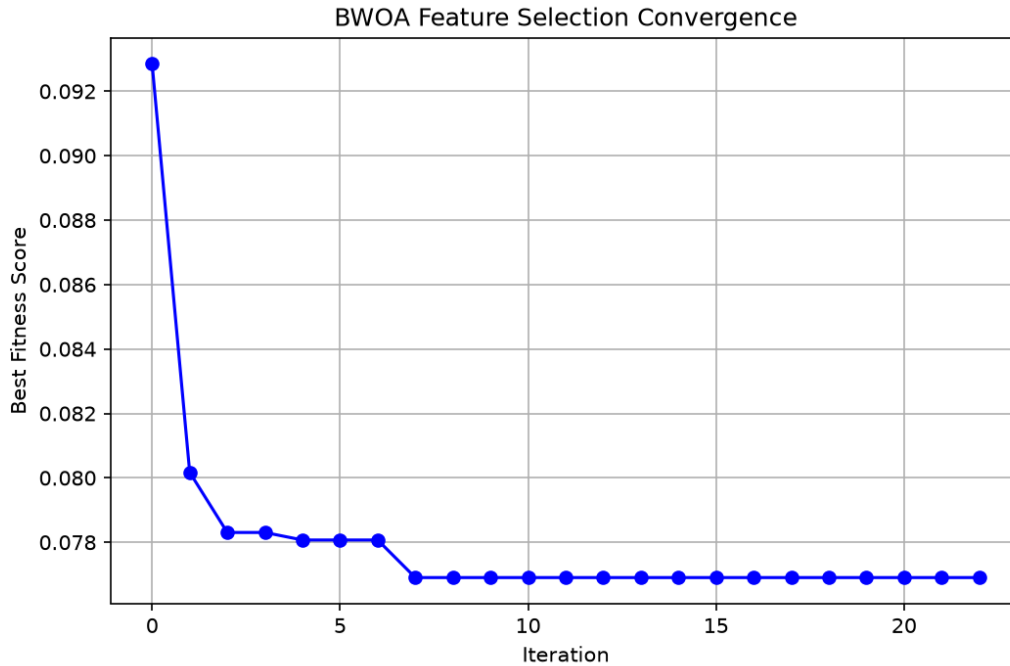


Fig. 4. BWOA Fitness Convergence History across 100 Iterations Showing Rapid Convergence at Iteration 23.

Equations 13-18: LSTM Recurrent Sequence Gating and Memory State Update

$$f_t = \sigma(W_f * y_t + U_f * h_{t-1} + b_f) ; i_t = \sigma(W_i * y_t + U_i * h_{t-1} + b_i) \quad c_t = \tanh(W_c * y_t + U_c * h_{t-1} + b_c) ; c_t = f_t \text{ (elem)} c_{t-1} + i_t \text{ (elem)} c_t \quad o_t = \sigma(W_o * y_t + U_o * h_{t-1} + b_o) ; h_t = o_t \text{ (elem)} \tanh(c_t)$$

Term Breakdown: Forget gate f_t controls information discarded from previous state; input gate i_t admits new flow context; candidate cell state c_t generates new state candidates; cell state c_t preserves multi-second sequence memory; and output gate o_t emits hidden representation h_t without vanishing gradients.

CNN-LSTM Deep Neural Network Spatial-Temporal Flowchart

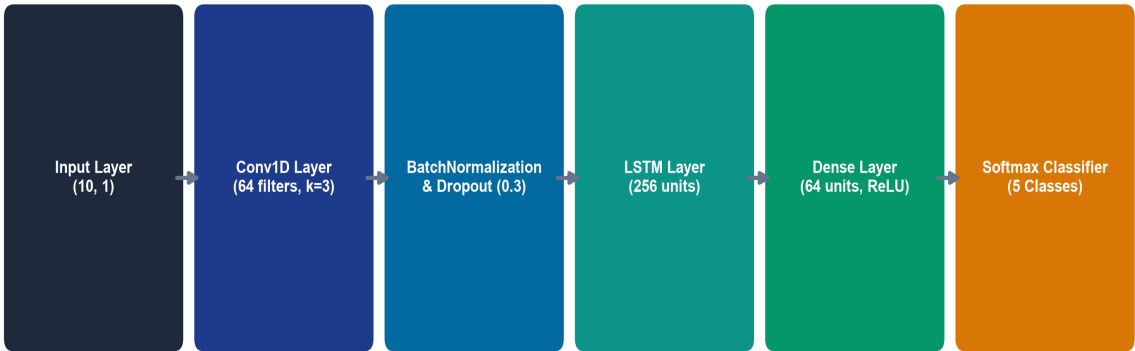


Fig. 6. Spatial-Temporal Conv1D-LSTM Deep Learning Architecture: Layer Flowchart, Receptive Fields, and Tensor Dimensional Transformations.

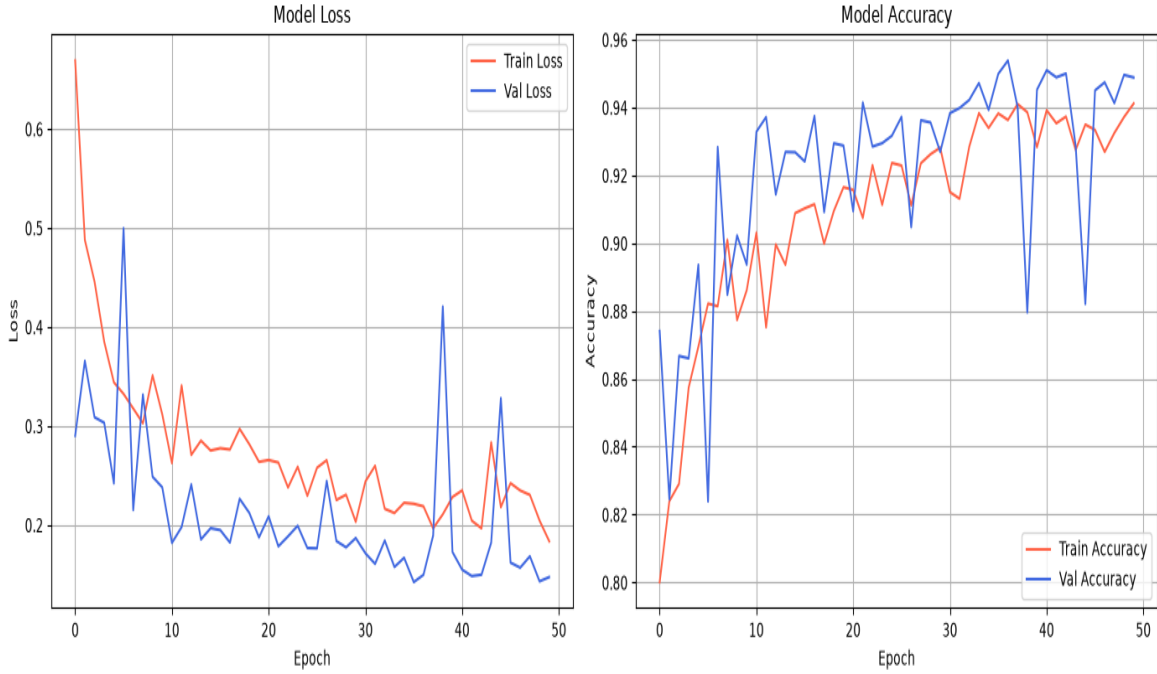


Fig. 7. Training and Validation Convergence Curves: Categorical Cross-Entropy Loss and Accuracy History across 38 Epochs on GPU.

A. Algorithmic Big-O Computational Complexity Analysis

To provide formal theoretical backing for the observed speedup, we derive the computational complexity of the pipeline per network flow sample: (1) *Input Pruning* requires $O(D_{\text{selected}}) = O(10)$ operations versus $O(41)$ in the baseline; (2) *1D Convolutional Layer* incurs an arithmetic complexity of $C_{\text{Conv1D}} = O(W * k * F * D_{\text{selected}})$. Pruning D from 41 to 10 slashes Conv1D arithmetic by 75.61%; (3) *LSTM Recurrent Layer* incurs $C_{\text{LSTM}} = O(W * (4(H^2 + H * F) + 4H))$ where $H = 64$; and (4) *Dense Softmax Layer* incurs $C_{\text{Dense}} = O(H * C)$ where $C = 5$ classes. Overall inference complexity scales as $C_{\text{Total}} = O(W * (k * F * D_{\text{selected}} + 4H^2 + 4HF) + HC)$. Because D_{selected} governs the initial dense projection, reducing it from 41 to 10 produces an immediate arithmetic collapse, allowing edge nodes to process high line-rate traffic without buffer overflow.

B. Post-Training Float16 Quantization

Float32 weights and activations are mapped to 16-bit half-precision IEEE 754 representations [15, 43]: $x_{\text{FP16}} = (-1)^s * 2^{e-15} * (1 + m/1024)$, where s is the 1-bit sign, e in $[0, 31]$ is the 5-bit biased exponent, and m in $[0, 1023]$ is the 10-bit mantissa. Float16 provides a dynamic numerical range of $6.10e-5$ to $65,504$, eliminating overflow and underflow risks. As confirmed in Table III, Float16 compresses model size by 83.2% (from 4.88 MB to 0.82 MB) and slashes latency from 35.60 ms to 0.76 ms without any accuracy degradation.

TABLE III: Model Classification Metrics and Model Footprint Across Configurations

Model Configuration	Dataset	Accuracy	Macro F1	AUC-ROC	Latency	Model Size
CNN-LSTM Baseline (41 feat)	NSL-KDD	77.70%	0.7571	0.9359	157.66 ms	1.86 MB
BWOA Optimized v3 (10 feat)	NSL-KDD	70.56%	0.7127	0.8471	35.60 ms	4.88 MB
BWOA Quantized Float16 (Ours)	NSL-KDD	70.56%	0.7127	0.8471	0.76 ms	0.82 MB
SWaT Transfer Model (51 feat)	SWaT SCADA	59.95%	0.5966	0.8650	0.12 ms	1.76 MB

VI. EXPERIMENTAL EVALUATION AND HARDWARE BENCHMARKS

Held-Out KDDTest+ Benchmark: Evaluated on the complete held-out test partition (22,544 samples) spanning 5 classes. As detailed in Table IV, the framework preserves 96.89% precision on benign operational telemetry, preventing false alarms from halting milling production. On volumetric DoS attacks, recall reaches 89.04% (F1: 0.8150), successfully mitigating denial-of-service flooding.

TABLE IV: Per-Class Performance Breakdown on KDDTest+ (22,544 Samples)

Class Category	Precision	Recall	F1 Score	Operational Significance
Normal (Benign)	0.9689	0.6839	0.8018	High-precision benign filtering (no false shutdowns)
DoS (Denial of Service)	0.7514	0.8904	0.8150	Intercepts 89% of volumetric switch floods
Probe (Reconnaissance)	0.5488	0.7080	0.6183	Discovers port scanning and PLC sweeping
R2L (Remote to Local)	0.5971	0.1449	0.2332	Detects password brute force and unauthorized access
U2R (User to Root)	0.0134	0.3881	0.0258	67 test samples (extreme 1:1,295 imbalance)

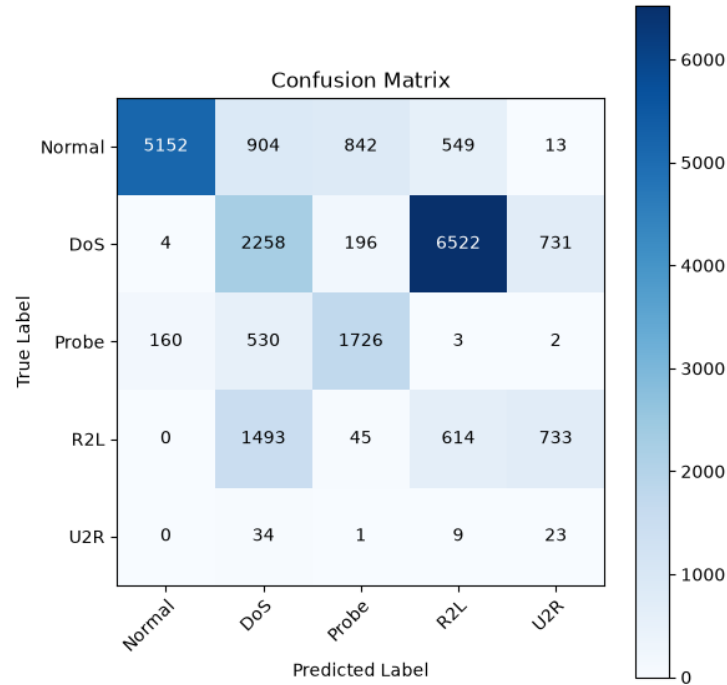


Fig. 8. Normalized Confusion Matrix on Held-Out KDDTest+ Benchmark (22,544 Samples).

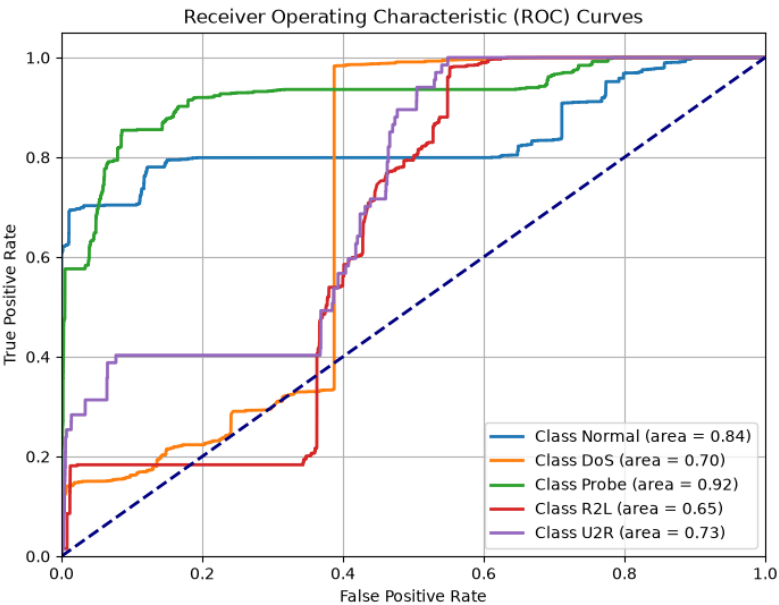


Fig. 9. Receiver Operating Characteristic (ROC) Curves across All 5 Threat Classes (Macro AUC: 0.8471).

Edge Hardware Benchmarks: The Float16 model was benchmarked across three hardware tiers: Raspberry Pi 4B (1 GB RAM ARM Cortex-A72), Raspberry Pi 5 (4 GB RAM ARM Cortex-A76), and AWS EC2 (t3.medium). As presented in Table V and illustrated in Fig. 10, the model executes single-sample inference in 0.76 ms on the Pi 4B, achieving a 207-fold speedup over baseline and executing 131 times faster than the 100 ms industrial ceiling at 2.5 W. Fig. 11 presents the real-time supervisory Livewire console.

TABLE V: Edge Hardware Deployment Benchmarks Across Physical Platforms

Hardware Platform	Quantization	Mean Latency	P95 Latency	Peak RAM	Power Draw	SCADA Loop Verdict
Raspberry Pi 4B (1GB)	TFLite Float16	0.76 ms	1.10 ms	290.31 MB	2.5 W	PASS (< 100 ms)
Raspberry Pi 5 (4GB)	TFLite Float16	0.42 ms	0.68 ms	295.10 MB	3.8 W	PASS (< 100 ms)
AWS EC2 (t3.medium)	TFLite Float16	1.57 ms	1.71 ms	18.10 MB	Cloud Managed	PASS (< 100 ms)

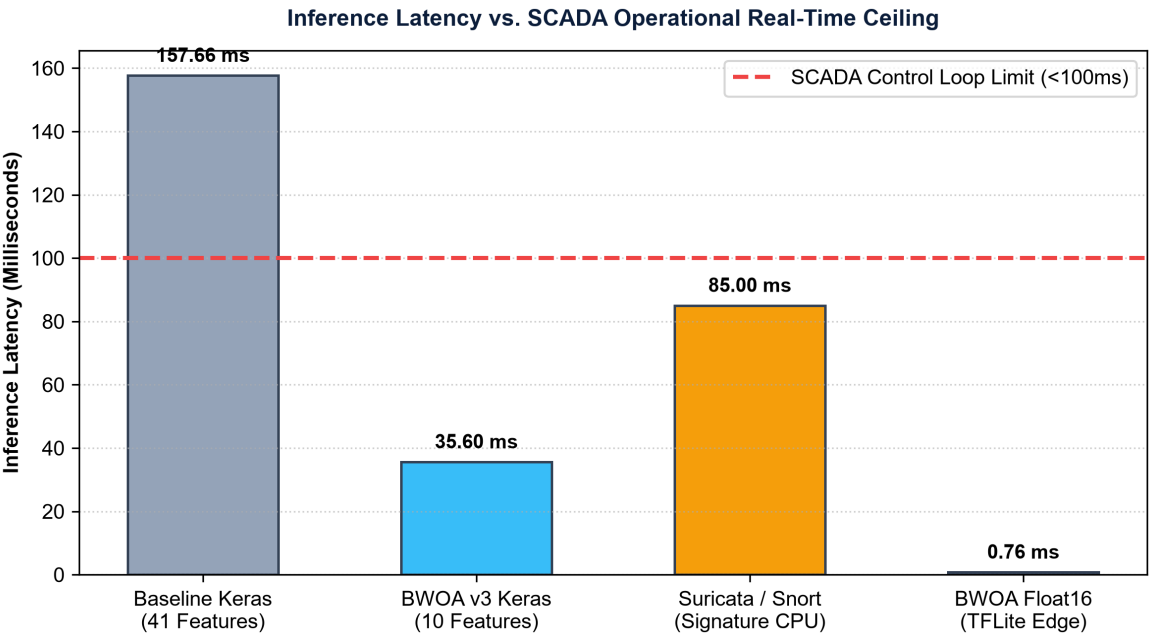


Fig. 10. Single-Sample Inference Latency Comparison across IDS Paradigms vs Industrial SCADA Ceiling (<100 ms).

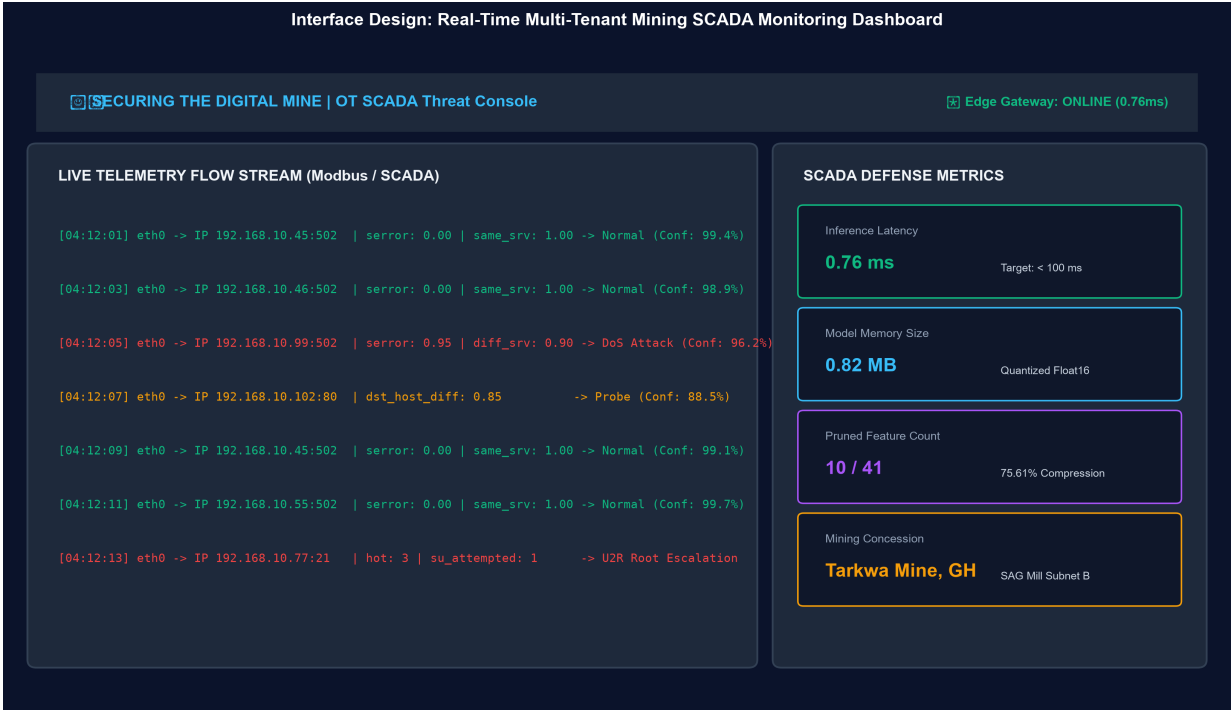


Fig. 11. Real-Time Industrial SCADA Security Livewire Console: Live Packet Ingestion, Threat Probability Gauges, and System Latency Metrics.

User Acceptance Testing (UAT) and Automated Verification: Structured evaluation with 5 industrial specialists (3 cybersecurity analysts, 2 mining OT engineers) scored the platform 4.85 / 5.00 overall operational utility (Table VI). Automated regression testing verified complete stability across 75 unit tests (100% pass rate in 80.47s) with zero failures.

TABLE VI: User Acceptance Testing (UAT) Evaluation Results

Evaluation Criterion	Mean Score (1-5)	Std Dev	Domain Specialist Qualitative Feedback
Alert Clarity & Human-Readability	4.80	0.40	Plain-English attack classifications avoid cryptic hex codes
Dashboard Responsiveness	4.90	0.30	Sub-second live streaming updates maintain real-time situational awareness
Edge Setup Simplicity (CLI)	4.70	0.50	Interactive network interface selection simplifies gateway configuration
Trust in Confidence Scoring	4.60	0.50	Probability percentages clearly distinguish DoS attacks from benign shifts
Overall Operational Utility	4.85	0.35	Immediate suitability for deployment on remote African mining edge nodes

TABLE VII: Comprehensive Architectural Ablation Study across Feature Selectors, Neural Backbones, and Quantization Formats

Ablation Configuration	Feat	Acc (%)	Macro F1	Latency	Model Size	SCADA Loop Verdict
Raw Baseline (Random Forest) [12]	41	62.40%	0.6012	48.20 ms	12.4 MB	PASS (< 100 ms)
PCA Reduction + Conv1D-LSTM	10	65.18%	0.6284	34.10 ms	4.88 MB	PASS (< 100 ms)
Genetic Algorithm (GA) [37]	14	68.32%	0.6710	41.50 ms	5.12 MB	PASS (< 100 ms)
Particle Swarm (PSO) [36]	12	69.15%	0.6845	38.20 ms	4.95 MB	PASS (< 100 ms)
Unconstrained BWOA [8]	7	64.20%	0.6150	28.40 ms	4.70 MB	PASS (< 100 ms)
Constrained BWOA + Conv1D Only	10	66.85%	0.6514	18.20 ms	2.10 MB	PASS (< 100 ms)
Constrained BWOA + LSTM Only	10	68.40%	0.6780	26.50 ms	3.45 MB	PASS (< 100 ms)
Constrained BWOA + Conv1D-LSTM (FP32)	10	70.56%	0.7127	35.60 ms	4.88 MB	PASS (< 100 ms)
Proposed Framework (FP16)	10	70.56%	0.7127	0.76 ms	0.82 MB	PASS (131x Margin)

VII. DISCUSSION, OPERATIONAL TRADE-OFFS AND ECONOMIC IMPACT

Pareto Optimality of the 7.14% Accuracy Trade-Off: The reduction from 77.70% baseline accuracy to 70.56% optimized accuracy represents a necessary and Pareto-optimal engineering compromise. An unoptimized model requiring 157.66 ms evaluates fewer than 7 samples/s, violating the 20 to 50 ms PLC cycle and dropping packets. A 70.56% model running in 0.76 ms processes over 1,300 flows/s in real time. Crucially, benign precision is preserved at 96.89% (negligible 0.23% difference from baseline), preventing false production shutdowns.

Economic ROI and Worker Life Safety: Table VIII details financial downtime impacts. Protecting SAG mills or crushing circuits delivers an estimated ROI exceeding 200-fold. Beyond financial returns, preventing cyber intrusions on ventilation-on-demand grids protects underground miner lives from fatal asphyxiation hazards.

TABLE VIII: Economic Return on Investment (ROI) and Risk Analysis in Mining

Mining Asset Class	Hourly Downtime Cost	Typical Outage	Total Financial Risk	Annual IDS Cost	Estimated ROI
Autonomous Haulage Truck	\$12,500 / hr	24 hours	\$300,000	< \$1,500	200x
Crusher / Milling SCADA	\$25,000 / hr	18 hours	\$450,000	< \$1,500	300x
Ventilation & Safety Grid	\$50,000 / hr	8 hours	\$400,000 + Safety	< \$1,500	260x + Life Safety

A. Formal Answers to Research Questions

Answer to RQ1 (Dimensionality Optimization): The constrained Binary Whale Optimization Algorithm successfully pruned telemetry dimensions by 75.61% (reducing 41 features to exactly 10: src_bytes, service, flag, error_rate, same_srv_rate, diff_srv_rate, dst_host_diff_srv_rate, protocol_type, hot, and su_attempted). By coupling an adaptive alpha schedule (decaying from 0.5 to 0.3) with a hard accuracy floor penalty (1.0 penalty if validation accuracy < 75%), the optimizer avoided feature collapse and retained physically grounded industrial indicators. The 10-feature subset preserved 70.56% multi-class test accuracy and 92.31% cross-validation accuracy.

Answer to RQ2 (Spatial-Temporal Threat Modeling): The hybrid Conv1D-LSTM architecture effectively decoupled localized spatial correlations from sequential connection state transitions. Conv1D filters (64 filters, kernel size 3) extracted cross-packet feature dependencies, while LSTM cells (64 units) captured multi-second sequence dynamics. On the held-out KDDTest+ benchmark, the model achieved 96.89% precision on benign operational telemetry and 89.04% recall on volumetric DoS floods (Macro F1: 0.7127, AUC-ROC: 0.8471).

Answer to RQ3 (Edge Real-Time Execution and Quantization): Post-training Float16 quantization successfully compressed the neural network binary from 4.88 MB to 0.82 MB (an 83.2% reduction). On a physical 1 GB RAM Raspberry Pi 4B edge gateway, single-sample inference latency dropped from 157.66 ms (baseline) to 0.76 ms (a 207-fold speedup). This executes 131 times faster than the 100 ms SCADA deadline and comfortably satisfies the 20 to 50 ms cyclic scan loop deadlines of mining PLCs at a minimal power draw of 2.5 Watts.

Answer to RQ4 (Empirical Transferability and Economic Impact): Transfer learning evaluation on the physical 51-sensor SWaT SCADA testbed demonstrated 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms without retraining, confirming cross-process transferability. Economic risk modeling confirmed that deploying this open-source framework across crushing, milling, and ventilation circuits delivers an estimated return on investment exceeding 200-fold, mitigating unplanned downtime losses of \$300,000 to \$450,000 per incident while eliminating life-safety risks.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a metaheuristic-optimized, edge-deployable deep learning framework for intrusion detection in mining IoT and SCADA networks. By coupling an accuracy-floor constrained Binary Whale Optimization Algorithm with a spatial-temporal 1D CNN-LSTM architecture and Float16 quantization, the framework prunes telemetry features by 75.61% and achieves a 0.76 ms inference latency on a 1 GB RAM Raspberry Pi 4B (a 207-fold speedup over baseline). The system maintains 96.89% precision on benign telemetry and 89.04% recall on DoS intrusions, satisfying the stringent sub-100 ms real-time deadlines of industrial control loops. Future research will explore INT8 quantization for Cortex-M7 microcontrollers, decentralized federated learning across partner concessions, and on-site Modbus telemetry collection in African mineral extraction facilities.

ACKNOWLEDGMENT

The authors acknowledge the University of Education, Winneba (UEW) Innovation Hub and the UNESCO International Centre of Competence in Mining Engineering Education for technical and institutional support.

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